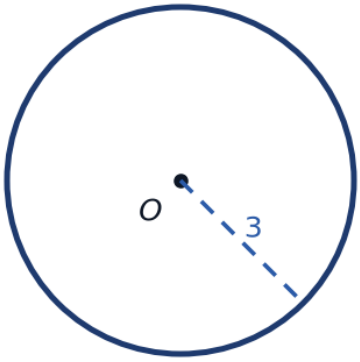
Vector-Valued Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

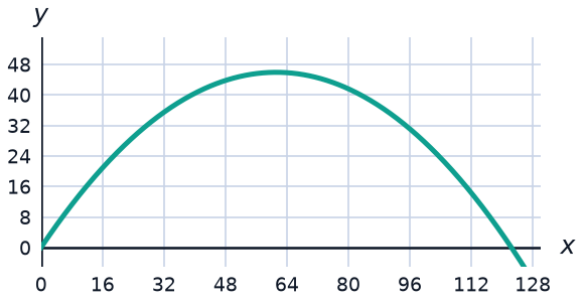
1 [Foundational] • Target: Motion on a circle and centripetal acceleration

A particle moves along $\vec{r}(t) = \langle 3\cos t, 3\sin t \rangle$. Find $\vec{v}(t)$, $\vec{a}(t)$, and verify that $\vec{a}(t)$ points toward the origin (is a scalar multiple of $-\vec{r}(t)$).



2 [Foundational] • Target: Projectile motion as a vector-valued function

A projectile is launched with position $\vec{r}(t) = \langle 20t, 30t - 4.9t^2 \rangle$ (meters, seconds). Find the velocity vector $\vec{v}(t)$ and the time when the projectile reaches maximum height (where the vertical velocity component is 0).



3 [Foundational] • Target: Position, velocity, and acceleration vectors

A particle's position is given by $\vec{r}(t) = \langle t^2, t^3 \rangle$. Find the velocity vector $\vec{v}(t)$ and acceleration vector $\vec{a}(t)$.

4 [Foundational] • Target: Position, velocity, and acceleration vectors

For the same particle, find the speed at $t = 2$.

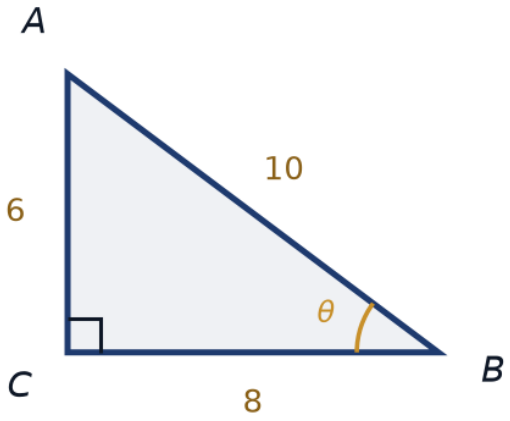
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5 [Foundational] • Target: Integrating vector-valued functions

Given $\vec{v}(t) = \langle 3t^2, 2t \rangle$ and initial position $\vec{r}(0) = \langle 1, 4 \rangle$, find $\vec{r}(t)$.

6 [Foundational] • Target: Total distance traveled

A particle has velocity $\vec{v}(t) = \langle \cos t, \sin t \rangle$ for $t \in [0, \pi]$. Find the total distance traveled.



7 [Foundational] • Target: Vectors and parametric motion combined

A particle moves along $x(t) = e^t$, $y(t) = e^{-t}$. Find the velocity vector and speed at $t = 0$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Vectors and parametric motion combined

For the particle above, find the magnitude of the acceleration vector at $t = 0$.

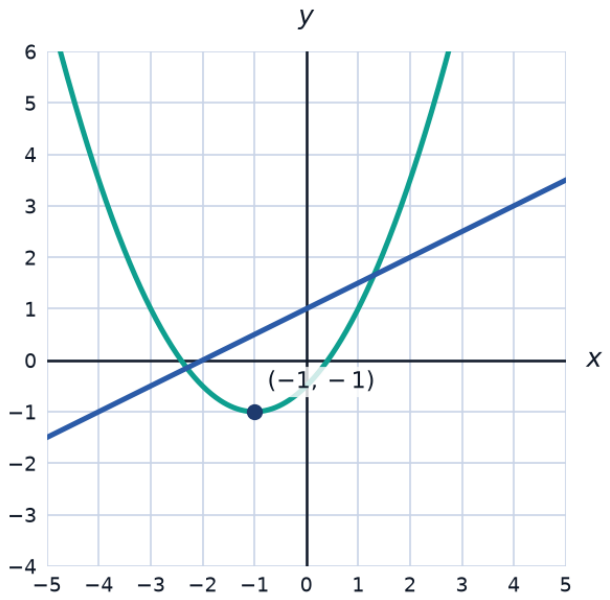
9 [Intermediate] • Target: Unit tangent vectors

For $\vec{r}(t) = \langle t, t^2 \rangle$, find the unit tangent vector $\vec{T}(t)$ at $t = 1$.

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10 [Intermediate] • Target: Unit tangent vectors

Explain why $\vec{T}(t) = \frac{\vec{v}(t)}{|\vec{v}(t)|}$ always has magnitude 1.



11 [Intermediate] • Target: Motion on a circle and centripetal acceleration

For the same particle, verify that $\vec{v}(t) \cdot \vec{r}(t) = 0$ for all t, and interpret this geometrically.

12 [Intermediate] • Target: Displacement vs. distance for vector motion

A particle has position $\vec{r}(t) = \langle t^2, t \rangle$ for $t \in [0, 2]$. Find the displacement vector from $t = 0$ to $t = 2$.

13 [Intermediate] • Target: Displacement vs. distance for vector motion

Set up (but do not evaluate) the integral for the total distance traveled by the particle $\vec{r}(t) = \langle t^2, t \rangle$ over $t \in [0, 2]$.

14 [Intermediate] • Target: Vector operations review

Given $\vec{u} = \langle 3, -2 \rangle$ and $\vec{w} = \langle -1, 4 \rangle$, find $2\vec{u} - 3\vec{w}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

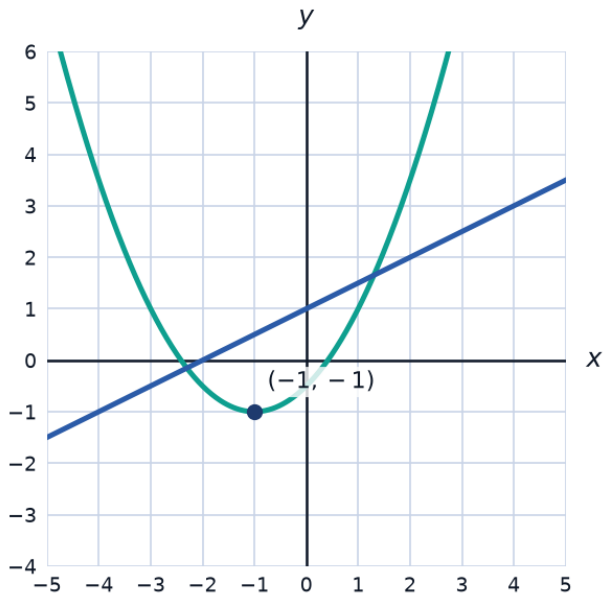
15 [Challenging] • Target: Vector operations review • Cross-Lesson: 6.01 Power Series and Interval of Convergence

Find the dot product $\vec{u} \cdot \vec{w}$ for $\vec{u} = \langle 3, -2 \rangle$ and $\vec{w} = \langle -1, 4 \rangle$, and state whether the vectors are perpendicular.

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16 [Challenging] • Target: Curvature-related speed problems • Cross-Lesson: 6.02 Taylor and Maclaurin Series

A particle moves along $\vec{r}(t) = \langle t^3, t^2 \rangle$. Find the times t when the particle's speed is exactly $\sqrt{13}$.



17 [Challenging] • Target: Curvature-related speed problems • Cross-Lesson: 1.01 Integration by Parts

For the particle $\vec{r}(t) = \langle t, \ln t \rangle$ $(t > 0)$, find the velocity vector and speed at $t = 1$.

18 [Challenging] • Target: Projectile motion as a vector-valued function • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

For the same projectile, find the speed at $t = 0$ (launch) and interpret its two components.

19 [Challenging] • Target: Finding when a particle is at rest • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

A particle moves along $\vec{r}(t) = \langle t^3 - 3t,\ t^2 - 4 \rangle$. Find all times t when the particle is momentarily at rest (both velocity components equal 0 simultaneously).

20 [Challenging] • Target: Finding when a particle is at rest • Cross-Lesson: 3.01 Parametric Equations

For the same particle, find the times when only the horizontal velocity component is 0.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Position, velocity, and acceleration vectors

Given $\vec{u} = \langle 7, -1 \rangle$ and $\vec{w} = \langle 3, 8 \rangle$, find $2\vec{u} + 3\vec{w}$.

a. $\langle 23, 22 \rangle$

b. $\langle 24, 22 \rangle$

c. $\langle 27, 13 \rangle$

d. $\langle 23, -22 \rangle$

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22 [Foundational] • Target: Integrating vector-valued functions

Given $\vec{u} = \langle 7, 7 \rangle$ and $\vec{w} = \langle 2, 8 \rangle$, find $-2\vec{u} + 3\vec{w}$.

a. $\langle -7, 10 \rangle$

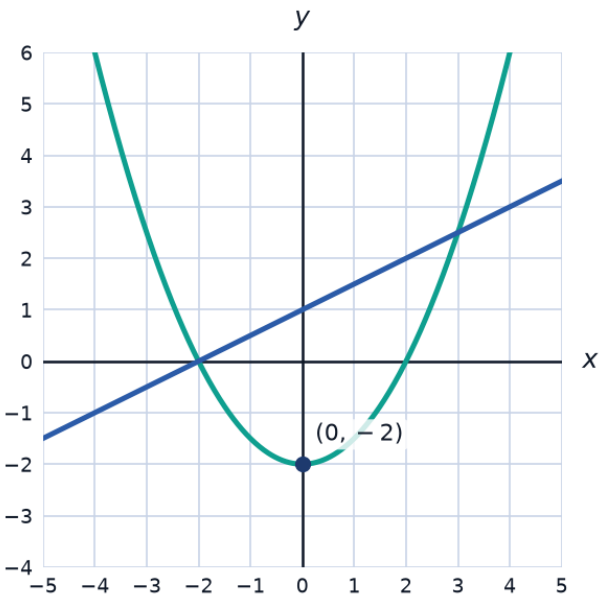
b. $\langle 17, 5 \rangle$

c. $\langle -8, -10 \rangle$

d. $\langle -8, 10 \rangle$

23 [Foundational] • Target: Total distance traveled

Given $\vec{u} = \langle 7, 3 \rangle$ and $\vec{w} = \langle 0, 6 \rangle$, find $2\vec{u} - 2\vec{w}$.



a. $\langle 14, -6 \rangle$

b. $\langle 14, 6 \rangle$

c. $\langle 15, -6 \rangle$

d. $\langle -14, 6 \rangle$

24 [Foundational] • Target: Vectors and parametric motion combined

Given $\vec{u} = \langle -7, -2 \rangle$ and $\vec{w} = \langle 8, -3 \rangle$, find $3\vec{u} - 2\vec{w}$.

a. $\langle -37, 0 \rangle$

b. $\langle -39, 1 \rangle$

c. $\langle -36, 0 \rangle$

d. $\langle 38, -5 \rangle$

25 [Foundational] • Target: Unit tangent vectors

Given $\vec{u} = \langle 6, -7 \rangle$ and $\vec{w} = \langle -7, -7 \rangle$, find $2\vec{u} + 3\vec{w}$.

a. $\langle -9, 35 \rangle$

b. $\langle -9, -35 \rangle$

c. $\langle -8, -35 \rangle$

d. $\langle 4, -35 \rangle$

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26 [Foundational] • Target: Motion on a circle and centripetal acceleration

Given $\vec{u} = \langle 1, 3 \rangle$ and $\vec{w} = \langle 5, 4 \rangle$, find $2\vec{u} + 3\vec{w}$.

a. $\langle 17, -18 \rangle$

b. $\langle 13, 17 \rangle$

c. $\langle 18, 18 \rangle$

d. $\langle 17, 18 \rangle$

27 [Foundational] • Target: Displacement vs. distance for vector motion

Given $\vec{u} = \langle 8, 0 \rangle$ and $\vec{w} = \langle 3, -1 \rangle$, find $2\vec{u} - 3\vec{w}$.

a. $\langle -18, -2 \rangle$

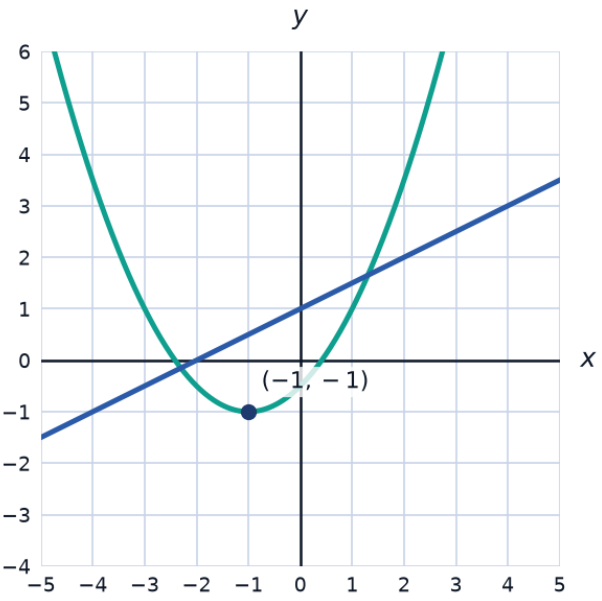
b. $\langle 7, -3 \rangle$

c. $\langle 7, 3 \rangle$

d. $\langle 8, 3 \rangle$

28 [Foundational] • Target: Vector operations review

A particle moves along $\vec{r}(t) = \langle 5t^2, 12t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.



a. 26

b. 28

c. 34

d. 13

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Curvature-related speed problems**

A particle moves along $\vec{r}(t) = \langle -8t^2, 15t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 36

b. 34

c. 46

d. 17

30 **[Intermediate]** • **Target: Projectile motion as a vector-valued function**

A particle moves along $\vec{r}(t) = \langle -24t^2, 7t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 62

b. 50

c. 52

d. 25

31 **[Intermediate]** • **Target: Finding when a particle is at rest**

A particle moves along $\vec{r}(t) = \langle -9t^2, -40t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 98

b. 84

c. 82

d. 41

32 **[Intermediate]** • **Target: Vector-valued functions and arc length**

A particle moves along $\vec{r}(t) = \langle 6t^2, -8t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 22

b. 20

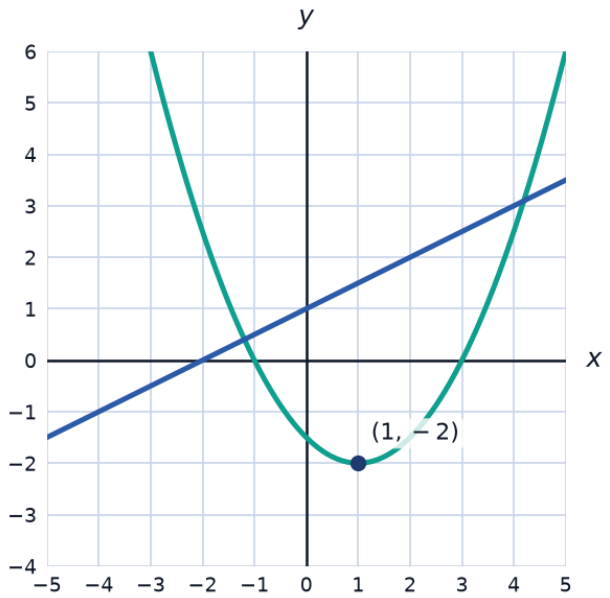
c. 28

d. 10

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33 **[Intermediate]** • **Target: Position, Velocity, and Acceleration Vectors**

A particle moves along $\vec{r}(t) = \langle 7t^2, 24t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.



a. 62

b. 25

c. 52

d. 50

34 **[Intermediate]** • **Target: Finding Position from Velocity & Acceleration**

A particle moves along $\vec{r}(t) = \langle -5t^2, 12t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 13

b. 26

c. 34

d. 28

35 **[Intermediate]** • **Target: Speed, Direction, and Displacement**

A particle moves along $\vec{r}(t) = \langle t^3 - 108t, t^2 + 3 \rangle$. Find the positive time t when the horizontal velocity component is 0.

a. 6

b. 18

c. 7

d. -6

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36 **[Intermediate]** • **Target: Distance Traveled with Vector-Valued Functions**

A particle moves along $\vec{r}(t) = \langle t^3 - 75t, t^2 + 3 \rangle$. Find the positive time t when the horizontal velocity component is 0.

a. 5

b. 6

c. -5

d. 15

37 **[Intermediate]** • **Target: Position, velocity, and acceleration vectors**

A particle moves along $\vec{r}(t) = \langle 4t, 6t - 3t^2 \rangle$. Find the time t when the vertical velocity component is 0 (the time of maximum height).

a. 1

b. 2

c. 3

d. 4

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Integrating vector-valued functions** • **Cross-Lesson: 1.02 Partial Fractions and Improper Integrals**

A particle moves along $\vec{r}(t) = \langle 6t, 36t - 3t^2 \rangle$. Find the time t when the vertical velocity component is 0 (the time of maximum height).

a. 12

b. 7

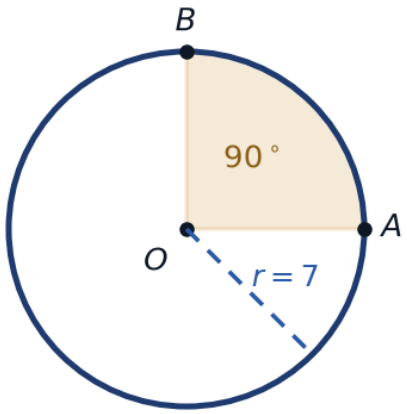
c. 5

d. 6

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39 [Challenging] • Target: Total distance traveled • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Find the arc length of $\vec{r}(t) = \langle 8t, 15t \rangle$ for $t \in [0, 7]$.



a. 119

b. 136

c. 161

d. 126

40 [Challenging] • Target: Vectors and parametric motion combined • Cross-Lesson: 3.01 Parametric Equations

Find the arc length of $\vec{r}(t) = \langle 9t, 40t \rangle$ for $t \in [0, 4]$.

a. 164

b. 205

c. 196

d. 168

41 [Challenging] • Target: Position, Velocity, and Acceleration Vectors • Cross-Lesson: 4.01 Polar Curves and Derivatives

A particle has position vector $\mathbf{r}(t) = \langle t^3 - 2t, \sin(2t) \rangle$. Find the acceleration vector $\mathbf{a}(t)$ at $t = 0$.

a. $\langle 0, 0 \rangle$

b. $\langle -2, 2 \rangle$

c. $\langle 0, -4 \rangle$

d. $\langle 6, 0 \rangle$

42 [Challenging] • Target: Position, Velocity, and Acceleration Vectors • Cross-Lesson: 4.02 Area in Polar Coordinates

What is the magnitude of the acceleration vector $\mathbf{a}(t) = \langle 3, -4 \rangle$?

a. 5

b. 7

c. $\sqrt{7}$

d. 1

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43 [Challenging] • Target: Finding Position from Velocity & Acceleration • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

If a particle has $\mathbf{v}(t) = \langle \frac{1}{t+1}, 2e^{2t} \rangle$ and $\mathbf{r}(0) = \langle 3, 4 \rangle$, find the x-component $x(t)$.

a. $x(t) = \ln(t+1) + 3$

b. $x(t) = \ln(t+1)$

c. $x(t) = -\frac{1}{(t+1)^2} + 3$

d. $x(t) = e^{t+1} + 2$

44 [Challenging] • Target: Speed, Direction, and Displacement • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Which vector represents the net displacement of a particle from $t = 1$ to $t = 4$?

a. $\int_1^4 \mathbf{v}(t)dt$

b. $\int_1^4 ||\mathbf{v}(t)||dt$

c. $\mathbf{v}(4) - \mathbf{v}(1)$

d. $\mathbf{a}(4) - \mathbf{a}(1)$

45 [Challenging] • Target: Distance Traveled with Vector-Valued Functions • Cross-Lesson: 6.01 Power Series and Interval of Convergence

Write an integral for the total distance traveled by a particle from $t = 0$ to $t = 5$ with $\mathbf{v}(t) = \langle \cos(t^2), e^{\sqrt{t}} \rangle$.

a. $\int_0^5 \sqrt{\cos^2(t^2) + e^{2\sqrt{t}}}\, dt$

b. $\int_0^5 (\cos(t^2) + e^{\sqrt{t}})dt$

c. $\sqrt{\int_0^5 \cos^2(t^2)dt + \int_0^5 e^{2\sqrt{t}}dt}$

d. $\int_0^5 ||\mathbf{a}(t)||dt$

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